

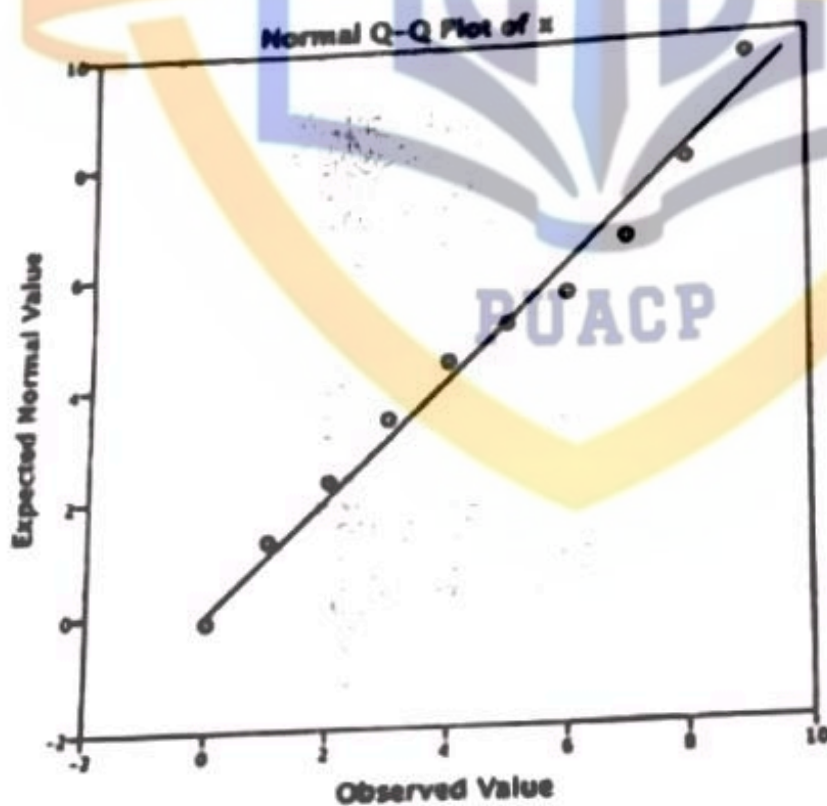


THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

(6×5=30)

Q.1. Answer the following short questions:

- Differentiate between categorical and quantitative variables. Describe the procedure of entering each type of variables in SPSS.
- Briefly explain how the SPSS data files are split and merged?
- How the date and weights variables are defined in SPSS? Give examples.
- Describe the procedure of creating a time series in SPSS.
- How missing values are treated in SPSS? Give examples.
- Interpret the plot given below.



Answers the following questions.

- Q.2: During the spring semester 2020 a team of Statistics randomly sampled 35 university students to study grade point average (GPAs). The following table gives the data.

Grade Point Average

2.2	3.4	4.0	3.1	3.3	3.8	3.0
2.8	2.6	3.1	3.2	3.0	2.8	3.5
3.2	3.0	2.5	2.5	2.8	3.4	2.2
2.7	2.7	2.6	2.9	3.0	2.8	2.9
2.9	2.6	3.5	2.5	2.5	2.5	3.5

Construct and interpret 90% and 95% confidence intervals for mean. Interpret the confidence intervals. Compare the lengths of the two intervals. (08)

- Q.3: Two varieties of tomato were experimented concerning their fruit-producing abilities, measures in pounds. The following data were obtained:

Location	1	2	3	4	5	6	7	8	9	10
Variety A	3.03	3.10	2.35	3.86	3.91	1.72	2.65	2.30	2.70	3.60
Variety B	2.28	3.63	2.17	3.56	3.73	1.85	1.48	1.86	2.76	2.68

Apply (i) The sign (ii) The Wilcoxon signed-rank tests to test the hypothesis that there is no difference in fruit-producing abilities of the two varieties. Use 5% level of significance. (10)

- Q.4: The following data were collected which are related to the length of an infant X_1 (cm), age X_2 (days), and weight X_3 (kg) (12)

X_1	57.5	52.8	61.3	67.0	53.5	62.7	56.2	68.5	69.2
X_2	78	69	77	88	67	80	74	94	102
X_3	2.75	2.15	4.41	5.52	3.21	4.32	2.31	4.30	3.71

- Compute all simple correlation coefficients and interpret the results.
- Calculate all partial correlation coefficients and interpret the results.
- Fit a multiple linear regression line X_1 on X_2 and X_3 .
- Test the overall significance of the above regression model using ANOVA table at $\alpha = 0.05$.